

Martin Birkenbaum

SENIOR SOFTWARE ENGINEER

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Senior Software Engineer with 8+ years of experience designing, building, and deploying scalable, high-performance backend systems and distributed microservices for cloud operations and fleet intelligence. Expertise in telemetry ingestion, operational automation, and GitOps-driven infrastructure, with a passion for reliability, observability, and developer toolchains.

SKILLS

- Languages:** Python, TypeScript, JavaScript, SQL, HTML/CSS
- Technologies & Frameworks:** Docker, Kubernetes, Helm, Git, GitOps, Distributed Systems, Cloud Infrastructure, AWS (S3, SQS), PostgreSQL (Schema Design, Query Tuning), Telemetry & Monitoring, Observability, Kafka/MSK, Jinja2, WeasyPrint, pytest

WORK EXPERIENCE

Senior Software Engineer at **Initech Corporation** 2021-03-01 – 2026-07-01

Led architecture and development of high-throughput microservices and GitOps-driven infrastructure pipelines.

- Designed and implemented Python-based event-driven microservices, processing high-volume telemetry and operational events (10M+ daily), contributing to fleet intelligence and operational insights.
- Standardized and optimized containerized deployments across cloud environments utilizing Docker, Kubernetes, Helm, and GitOps workflows, enhancing deployment quality and operational automation.
- Mentored 6 junior/mid-level engineers, establishing local code styling and robust testing standards to ensure service reliability.
- Developed custom internal toolchains that reduced CI pipeline execution times by 40%, improving developer productivity and observability into system health.

Software Engineer at **Hooli Inc.** 2018-06-01 – 2021-02-01

Backend engineer focused on building RESTful APIs, developer tooling, and automated deployment integrations.

- Developed robust customer-facing and agent-facing REST APIs in Python/Django and TypeScript/Node.js, enabling diverse backend services.
- Integrated third-party APIs and automated webhook processors for external developer platforms, supporting high-volume ingestion and real-time operational workflows.
- Implemented a suite of unit and integration tests raising coverage from 60% to 92%, significantly improving service reliability and quality.

PROJECTS

JobGitOps (Python, Nix, GitHub Actions) 2026-06-01 – 2026-07-01

A GitOps-driven job application and tracking system utilizing GitHub Issues, Projects, and Actions.

- Developed python-jobspy scraper integration and two-pass LLM fit triage
- Designed reproducible local environments via Nix and devenv

EDUCATION

Bachelor of Science in Computer Science at **University of Washington**

2014-09-01 – 2018-06-01

GPA: 3.8